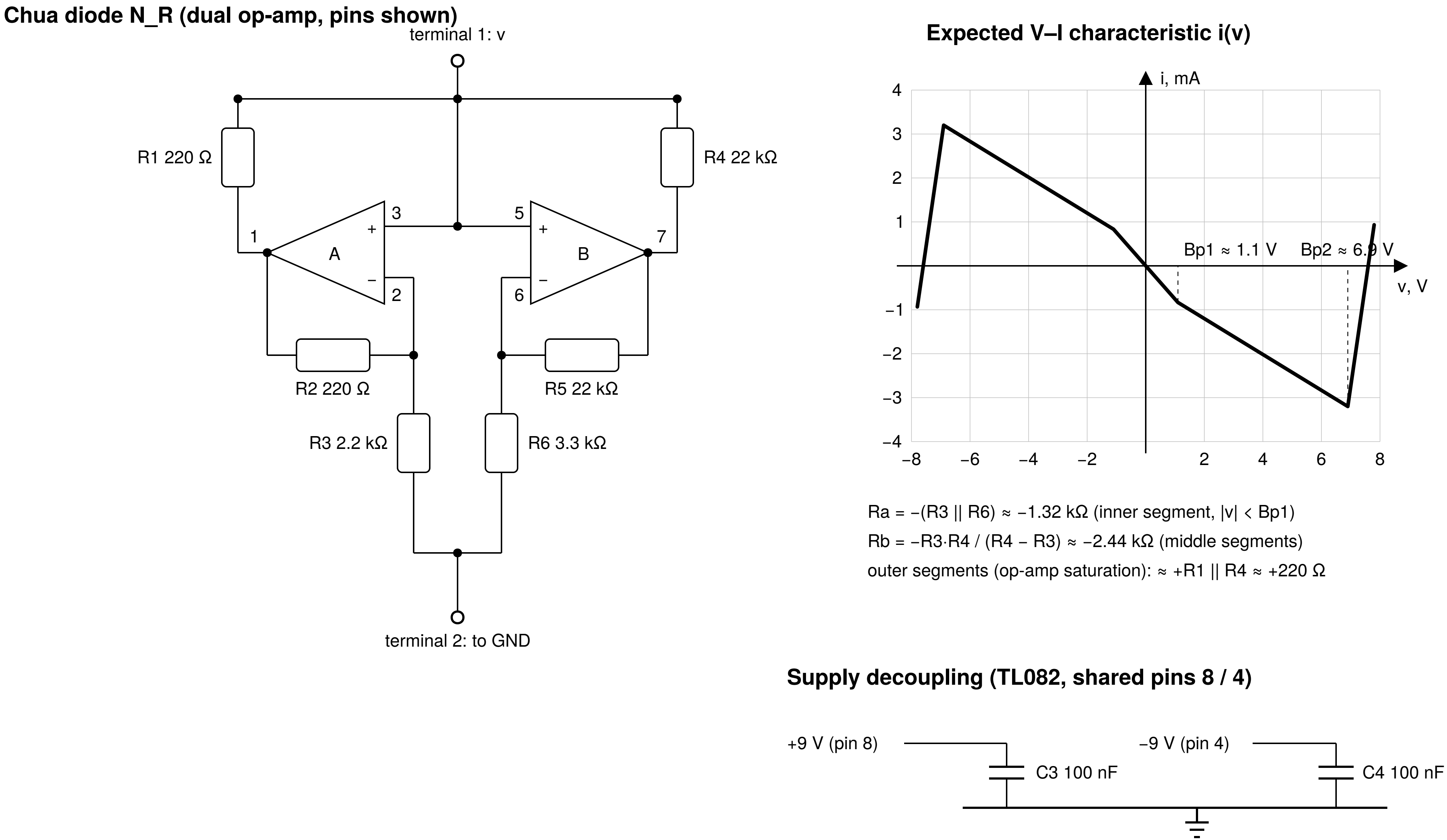


Fig. 1. Chua diode as a two-terminal element: circuit, expected piecewise-linear V–I curve, and supply decoupling.

## Chua oscillator ($N_R$ shown as a two-terminal element)

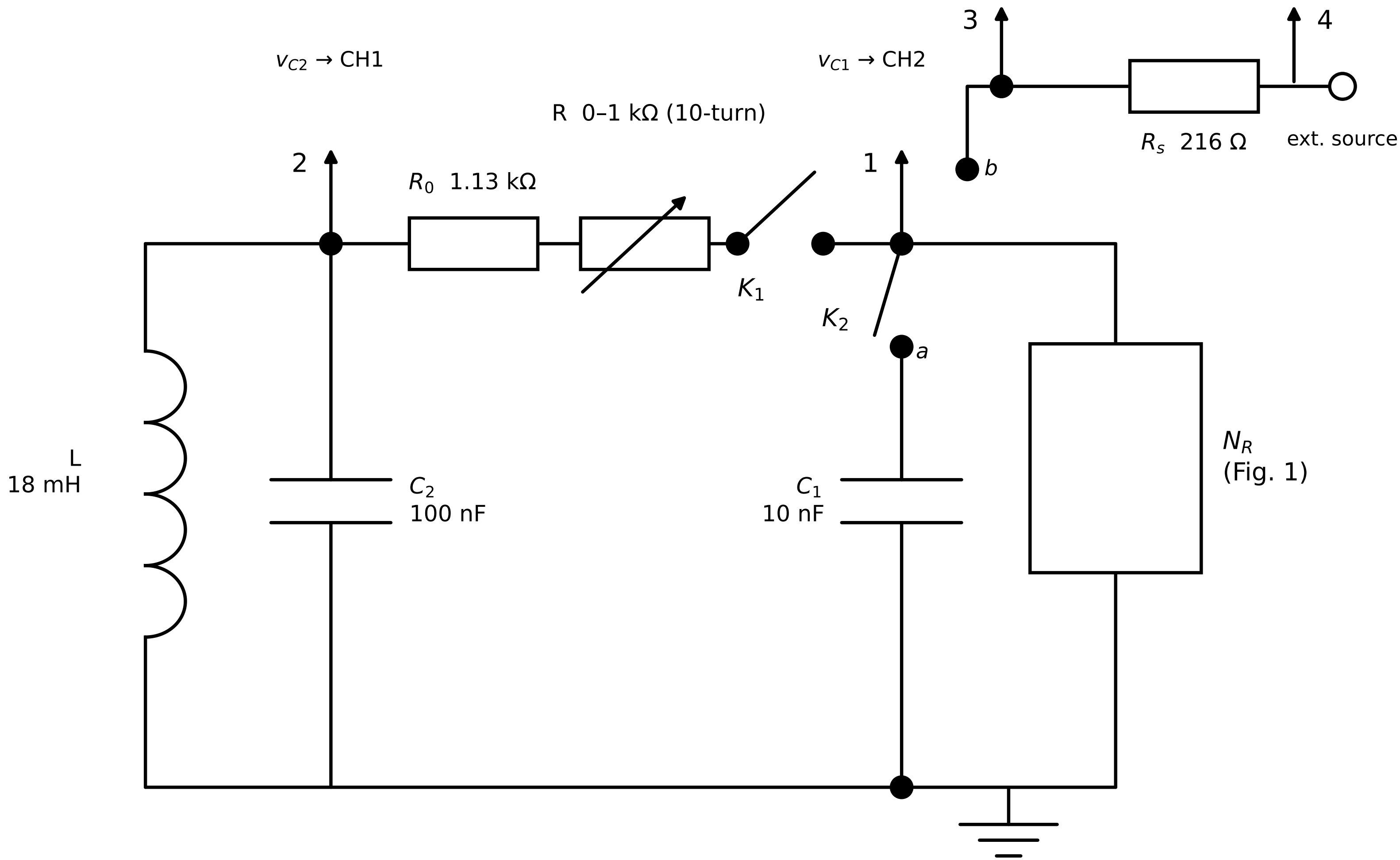


Fig. 2. Chua oscillator: resonant part (L, $C_2$), coupling resistance $R_0$ + R with disconnect switch $K_1$, $C_1$, and the nonlinear element $N_R$ of Fig. 1. $K_2$ in position a — oscillator mode; position b — V–I measurement of $N_R$ (with $K_1$ open): a slow triangular sweep is applied at terminal 4, $v = V_3$ is the voltage across $N_R$ and $i = (V_4 - V_3)/R_s$ its current.
Probe points: 1 = $v_{C1}$, 2 = $v_{C2}$, 3 and 4 across the shunt $R_s$. Coupling resistance $R_{tot} = R_0 + R$ = 1.13…2.13 kΩ; measurement points 1.54 / 1.74 / 1.91 kΩ. Dimensionless parameters: $\alpha = C_2/C_1 = 10$, $\beta = R_{tot}^2 C_2/L$ = 13…20.